

Model Card: TrustRAG

A self-verifying retrieval-augmented question answering system

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Model Details

TrustRAG is a pipeline built from three pretrained open-source models. No model was trained or fine-tuned for this project. The contribution is the way the models are combined: an answer is generated from retrieved text and then checked against that text before it is shown to the user.

Role	Model	Size	License (confirm on the model page)
Embeddings	sentence-transformers/all-MiniLM-L6-v2	about 22M parameters, 384-dimensional vectors	Apache 2.0
Answer generation	google/flan-t5-base	about 250M parameters	Apache 2.0
Verification (NLI)	roberta-large-mnli	about 355M parameters	MIT

Supporting software: FAISS (vector search), HuggingFace Transformers, Streamlit (interface), PyPDF2 and python-docx (file upload).

Pipeline settings used: chunk size 500 characters, overlap 50, top-k 3 retrieved chunks, generation limited to 100 tokens, generator and verifier inputs truncated to 512 tokens.

Output: an answer, a verdict (Supported or Unsupported), a confidence score, and the source chunk that supports the answer. An answer is Supported when at least one retrieved chunk is classed as entailing it. Each chunk is checked separately, because checking all chunks joined into one block produced a false contradiction during testing.

Version: 1.0 (academic project, October 2026)

Intended Use

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- Asking questions about a single contract, policy or similar document and seeing whether each answer is backed by the text.
- Demonstrating hallucination detection in retrieval-augmented generation for teaching and research.
- Exploring how retrieval settings such as chunk size affect answer support.

Intended users: students, researchers and reviewers who want a quick first read of a document and a signal about how far to trust each answer.

Out-of-scope uses

- Legal, medical, insurance or financial advice, or any decision where the answer is not checked against the original document by a person.
- Documents that are scanned images, because no OCR is included.
- Non-English documents, which were not tested.
- Automated decision making without human review.

Factors

- **Document type:** tested on English commercial legal contracts. Other genres, such as insurance policies or HR handbooks, were not evaluated.
- **Question phrasing:** long or instruction-style questions made the small generator echo the question instead of answering.
- **Document length and structure:** the verifier reads about 512 tokens at a time, and the answer must be supported within a single chunk.
- **Retrieval settings:** chunk size, overlap and k change what evidence the verifier sees.

Metrics

- **Support rate:** the share of generated answers labelled Supported.
- **Verifier confidence:** the NLI probability for the predicted class (entailment, neutral or contradiction).
- **False-claim rejection:** whether a deliberately false statement is rejected by every retrieved chunk.

Not measured: agreement with the annotated CUAD answers, and the verifier's precision and recall.

Evaluation Data

CUAD (Contract Understanding Atticus Dataset) from The Atticus Project, licensed CC BY 4.0. The version used has 22,450 training rows and 4,182 test rows. Evaluation used contracts from the training split, each contract being one document.

Training Data

None of the three components was trained by the author. Their training data is described on their own model pages (MiniLM and Flan-T5 on large general text and instruction collections, RoBERTa-large-MNLI on the MultiNLI inference corpus).

Quantitative Analyses

Test	Result
Electric City distributor agreement, 15 questions	15 of 15 Supported (100%)
BlueRock agency agreement, 2 unique questions	1 Supported, 1 Unsupported
Fabricated claim naming a different company and year	Rejected by all chunks; strongest chunk gave contradiction at 93%
Joined chunks versus per-chunk check on one answer	Joined: contradiction (0.65). Per-chunk: entailment (0.77)
Assignment-rights question	Supported at 96.7%
Speed on Colab (observed)	about 8 seconds per question, 41 seconds for 5 questions

Add your final multi-contract figure and chunk-size comparison here after running the evaluation notebook.

Reading these numbers: the sample is small. A high support rate shows that answers were backed by retrieved text, not that they are correct. The one Unsupported case was a non-answer, where the generator echoed the question text, and the verifier correctly refused to approve it.

Ethical Considerations

- **Not professional advice.** Supported means the retrieved text entails the answer, not that the answer is legally or medically right.
- **Privacy.** Uploaded documents are processed inside the running Colab session and no external API is called. However, the demo is shared through a public tunnel link, so anyone holding the link can reach the app. Do not upload confidential or personal documents.
- **Bias and coverage.** CUAD is English-language commercial contracts, so behaviour on other languages, jurisdictions or document styles is unknown. The component models also inherit biases from their training data.
- **Over-trust.** A green Supported label can encourage people to skip reading the source. The interface shows the supporting text so that users can check it themselves.

Caveats and Recommendations

- Always read the supporting chunk before relying on an answer.
- Treat Unsupported as a prompt to check the document, not as proof that the answer is wrong.
- Rephrase long or instruction-style questions into short, direct ones.
- Retrieval can miss the right passage, in which case a correct answer may be marked Unsupported.
- Future improvements: compare answers with annotated gold answers, use a larger generator and a long-context verifier, check answers sentence by sentence, and add OCR.

How to Use

Run `TrustRAG_App_Colab.ipynb` in Google Colab and open the link printed in Step 6, or run `streamlit run app.py` locally after `pip install -r requirements.txt`. Use `TrustRAG_Evaluation.ipynb` to reproduce the evaluation.

Citation

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